

BILIM VA MALAKALARNI BAHOLASH AGENTLIGI DARAJASIDA

Bilimingga ishon va muvaffaqiyatga erish!

UMUMTA'LIM VA MAXSUS FANLARDAN BILISH DARAJASINI ANIQLASHNING MILLIY
SERTIFIKAT IMTIHONI ISHTIROKCHILARI UCHUN

MATEMATIKA
FANIDAN
TEST TOPSHIRIQLARI KITOBIDIR

Ishtirokchining familiyasi, ismi va otasining ismi

Imzo

Testni tuzuvchi: **TypoThesis jamoasi**

Telegram kanal: **@Milliymath_sertifikat**

Ushbu test varianti 45 ta (1-45) topshiriqdan iborat.

Test topshiriqlari RASCH MODEL asosida baholanadi.

Kitobning yopiq va ochiq turdagi test topshiriqlari mavjud.

- yopiq turdagi test topshiriqlarida javob variantlari (A, B, C va D) orasidan aynan bitta javobni tanlang va javoblar varaqasidagi topshiriq raqamiga mos qatorda belgilang.
- ochiq turdagi test topshiriqlarining javobini javoblar varaqasidagi topshiriq raqamiga mos qatorda aniq, tushunarli va eng sodda ko'rinishda yozing.
- moslashtirishni talab qiluvchi yopiq test topshiriqlari uchun umumiy oltita (A – F) javob variantlari berilgan, uchta topshiriqqa (33-35 — test topshiriqlariga) ushbu javoblar orasidan mos ravishda bittadan javob tanlang va javoblar varaqasiga belgilang.

MILLIY SERTIFIKAT NAMUNAVIY SAVOLLAR TO'PLAMI!
2.03.2026. 2-smena.

1. 2026^{2025} sonini 7 ga bo'lgandagi qoldiqni toping.

- A) 5
- B) 3
- C) 4
- D) 6

2. Quyidagilardan nechitasi ratsional son bo'ladi:

a) $\frac{22}{7}$, b) $\frac{7}{17}$, c) $2 + \lg 3$, d) $\sqrt[3]{1 + \sqrt{2}}$, e) $\sqrt{\ln e^4} + \sqrt{\sqrt[3]{8} - 1}$.

- A) 1 ta
- B) 2 ta
- C) 3 ta
- D) 4 ta

3. Ikki xonali sonning o'nlar xonasidagi raqami birlar xonasidagi raqamdan 3 ga katta. Ikki xonali sonning raqamlar yig'indisini ikki xonali songa ko'paytirilsa 364 ga teng bo'ladi. U holda bu sonning raqamlari ko'paytmasini toping.

- A) 7
- B) 12
- C) 8
- D) 10

4. O'quvchi 2 ta kitobni 205000 so'mga oldi. Agar birinchi kitob narxi 15% ga pasayib ikkinchisini 20% ko'tarilsa kitob narxlari teng bo'ladi. Birinchi kitob ikkinchisidan qancha qimmat?

- A) 35000
- B) 33000
- C) 36000
- D) 32000

5. Hisoblang:

$$\left(\left(5^{\frac{4}{7}} \right)^{\frac{7}{2}} + (16^3)^{\frac{1}{12}} \right) \cdot 3^{-3}.$$

- A) $\frac{25}{27}$
- B) 1
- C) $\frac{29}{27}$
- D) $\frac{10}{3}$

6. Hisoblang:

$$\frac{1}{\sqrt{2}+2} - \frac{1}{2-\sqrt{2}} - \sqrt{2}.$$

- A) $-\sqrt{2}$
- B) $-2\sqrt{2}$
- C) $2\sqrt{2}$
- D) $\sqrt{2}$

7. $x > 8$ uchun ifodani soddalashtiring:

$$\sqrt{x+4\sqrt{x-4}} + \sqrt{x-4\sqrt{x-4}}.$$

- A) $2\sqrt{x-4}$
- B) $2x$
- C) 4
- D) $\sqrt{x-4}$

8. Arifmetik progressiyaning n ta hadi o'rta arifmetigi $2n$ ga teng bo'lsa, a_1 ni toping.

- A) 36
- B) 42
- C) 40
- D) 38

9. Maxraji natural son bo'lgan geometrik progressiyaning hadlari uchun $b_1+b_2+b_3 = 21$ va $\frac{1}{b_1} + \frac{1}{b_2} + \frac{1}{b_3} = \frac{7}{12}$ bo'lsa b_2 ni toping.

- A) 3
- B) 6
- C) 9
- D) 12

10. a, b sonlar uchun $(a+b+2024)^2 = 4(a+2023)(b+1)$ bo'lsa, $a-b$ ni toping.

- A) -2024
- B) -2022
- C) -2025
- D) -2026

11. $(2\sqrt{11}; 7\sqrt{5})$ sonlar orasida nechta butun son bor?

- A) 8 ta
- B) 9 ta
- C) 10 ta
- D) 11 ta

12. Soddalashtiring:

$$\frac{1 - \cos^4 \alpha - \sin^4 \alpha}{\operatorname{tg}^2 \alpha} .$$

- A) $2 \cos^4 \alpha$
- B) $2 \cos^2 \alpha$
- C) $2 \sin^2 \alpha$
- D) $2 \sin^4 \alpha$

13. Tenglamaning haqiqiy ildizlari yig'indisini toping:

$$\left(\sqrt{7 + 4\sqrt{3}} \right)^x + \left(\sqrt{7 - 4\sqrt{3}} \right)^x = 14 .$$

- A) 1
- B) 2
- C) -4
- D) 0

14. Nechta butun son tengsizlikni yechimi bo'la oladi:

$$||2x - 1| - x| \leq 7 .$$

- A) 12 ta
- B) 11 ta
- C) 14 ta
- D) 13 ta

15. Tengsizlikni nechta butun yechimi bor:

$$|1 - \sqrt{x}| < 1 .$$

- A) 2 ta
- B) 4 ta
- C) 5 ta
- D) 3 ta

16. Tenglamaning haqiqiy ildizlari yig'indisini toping (Agar u bitta bo'lsa o'sha ildizini toping):

$$\log_{3x-7}(2x-3)^2 + \log_{2x-3}(6x^2 - 23x + 21) = 4 .$$

- A) 3
- B) $7\frac{3}{4}$
- C) 4
- D) $3\frac{3}{4}$

17. Tenglama eng katta musbat ildizini toping:

$$x^4 - 10x^3 + 26x^2 - 10x + 1 = 0 .$$

- A) 4
- B) 6
- C) $3 + 2\sqrt{2}$
- D) $2 + \sqrt{3}$

18. Tenglamaning haqiqiy ildizlar sonini toping:

$$\sqrt{4 - x^2} \cdot \cos x = 0 .$$

- A) 2 ta
- B) 4 ta
- C) 5 ta
- D) 3 ta

19. Tenglamaning nechta haqiqiy ildizi bor:

$$x^2 - (\pi - \sqrt{10}) \cdot |x| - \sqrt{10} \cdot \pi = 0 .$$

- A) 2 ta
- B) 4 ta
- C) 5 ta
- D) 3 ta

20. $|a| > 1$, $-1 < b < 1$, $-1 \leq c < 1$ sonlari uchun $f(x, y, z) = \frac{x^2 + 1}{x^2 - 1} + \frac{2y^2 - 1}{2y^2 + 1} + \frac{4z^2 - 1}{4z^2 + 1}$ va $(x, y, z) = \left(\frac{\sqrt{a^2 - 1}}{a - 1}; \sqrt{\frac{b^2 - 1}{2(b - 1)^2}}; \frac{\sqrt{c^2 + 1}}{2(c - 1)} \right)$ bo'lsa, $f(x, y, z)$ ni soddalashtiring.

- A) $a + b + c$
- B) 1
- C) $\frac{a + b + c}{2}$
- D) $a + b - c$

21. Agar $f(x) = \begin{cases} 8 + x, & x < 0 \\ 2 - x, & x \geq 0 \end{cases}$ bo'lsa, $f(f(f(-5)))$ ni toping.

- A) 6
- B) 3
- C) 7
- D) 5

22. $f(x) = |\ln x - \log_{\cos x} \sin x|$ ning $x_0 = \frac{\pi}{4}$ nuqtadagi hosilasini toping.

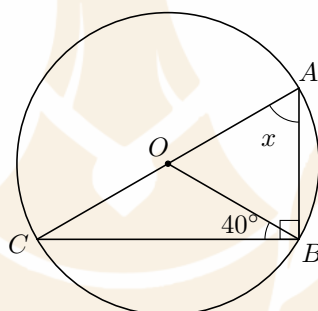
- A) $-\frac{4}{\pi} - \frac{4}{\ln 2}$
- B) $\frac{4}{\pi} - \frac{4}{\ln 2}$
- C) $-\frac{4}{\pi} + \frac{4}{\ln 2}$
- D) $\frac{4}{\pi} + \frac{4}{\ln 2}$

23. Hisoblang:

$$\int_0^1 (x^3 - x^2 + 2)^{2025} \cdot (3x^2 - 2x) dx .$$

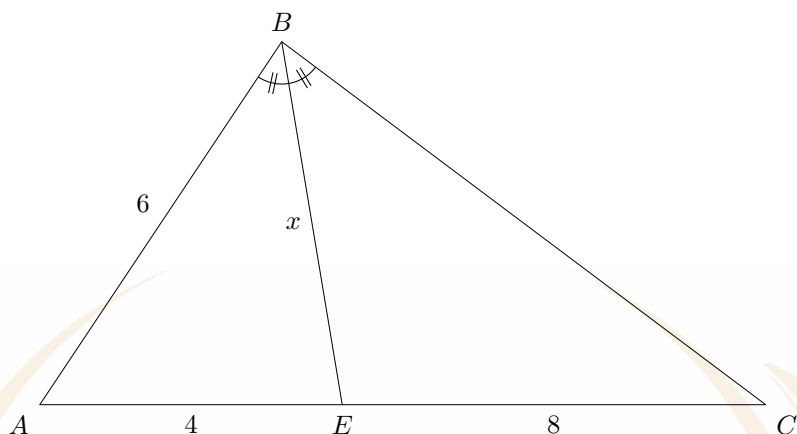
- A) 2025
- B) $\frac{2026}{2025}$
- C) 0
- D) $-\frac{1}{2025}$

24. Chizmada to'g'ri burchakli uchburchak va aylana tasvirlangan. x burchakni toping.

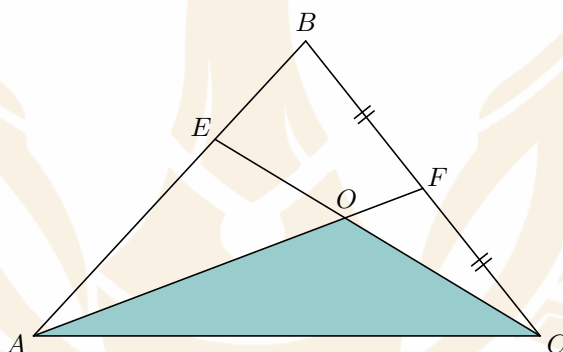


- A) 50°
- B) 60°
- C) 45°
- D) 40°

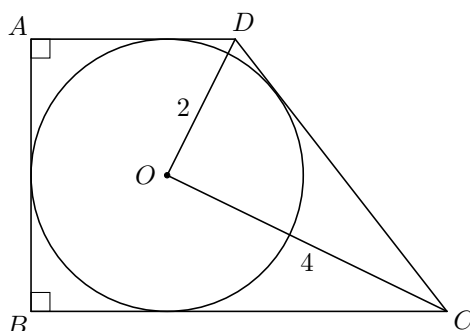
25. ABC uchburchakning BE bissektrisasi AC asosni 4 va 8 ga teng kesmalarga ajratadi. $AB = 6$ bo'lsa, $BE = x$ ni toping.



- A) $2\sqrt{10}$
 B) $\sqrt{10}$
 C) $2\sqrt{5}$
 D) $4\sqrt{10}$
26. ABC trapetsiyada AF medianasi. $EB : AB = 1 : 3$ va $S_{ABC} = 60$ bo'lsa, S_{AOC} ni toping.



- A) 12
 B) 24
 C) 36
 D) 18
27. $ABCD$ trapetsiyaga ichki aylana chizilgan. Aylana markazidan D va C uchlarigacha bo'lgan masofa 2 va 4 ga teng. Shu shart bilan trapetsiya yuzini toping.



- A) 14
- B) 15
- C) $\frac{36}{5}$
- D) $\frac{72}{5}$

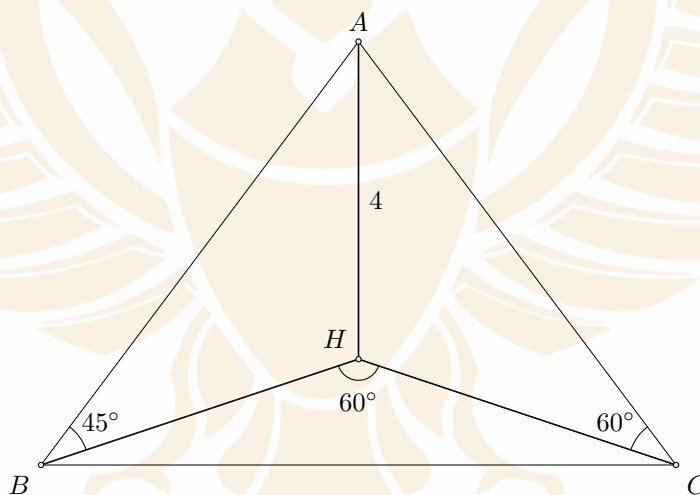
28. Muntazam oltiburchakning tomoni 1 ga teng va uning tomonlari o'rtalari tutashirildi va yana o'rtalari tutashtirildi. Bu ish cheksiz davom etdi. Hosil bo'lgan yuzalar yig'indisini toping.

- A) $12\sqrt{3}$
- B) $3\sqrt{3}$
- C) $4\sqrt{3}$
- D) $6\sqrt{3}$

29. $\vec{a} = (6; 2)$ va $\vec{b} = (4; -12)$ vektorlardan tashkil topgan uchurchakka tashqi chizilgan aylana radiusini toping.

- A) $10\sqrt{2}$
- B) 5
- C) $5\sqrt{2}$
- D) 10

30. $ABCH$ piramidaning AH qirrasi asos tekisligiga perpendikulyar va 4 ga teng. Chizmaga asosan piramida hajmini toping.

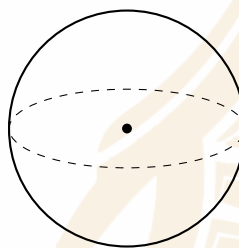
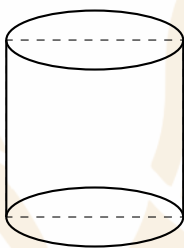


- A) 5
- B) 6
- C) $\frac{17}{3}$
- D) $\frac{16}{3}$

31. $A = \{1, 2, 3, 4, 5, 6\}$ to'plam berilgan. To'plamning nechta qism to'plamida 3 yoki 5 dan kamida bittasi qatnashadi.
- A) 64
B) 48
C) 32
D) 16
32. 1000 dan kichkina natural sonlar ichidan 11 ga bo'linadiganlari tanlab olindi. Ular ichida 2 ga va 3 ga bo'linmaydiganlari nechta?
- A) 90
B) 45
C) 30
D) 15

Topshiriqlar (33–35) va javob variant (A–F) larni o'zaro moslashtiring.

Hajmlari teng bo'lgan konus, silindr va shar berilgan. Konus o'q kesimi muntazam uchburchakdan, silindr o'q kesimi kvadratdan iborat bo'lsin.



33. $\frac{r_{\text{silindr}}}{r_{\text{konus}}}$ ni toping.

34. $\frac{r_{\text{shar}}}{r_{\text{silindr}}}$ ni toping.

35. $\frac{r_{\text{shar}}}{r_{\text{konus}}}$ ni toping.

Javoblar.

A) $\frac{\sqrt[6]{3}}{\sqrt[3]{6}}$

B) $\sqrt[3]{6}$

C) $\frac{\sqrt[6]{3}}{\sqrt[6]{4}}$

D) $\frac{\sqrt[3]{12}}{2}$

E) $\sqrt[3]{\frac{1}{4}}$

F) $\frac{\sqrt[6]{12}}{2}$

36. Tenglamani yeching:

$$2 \cdot (x^2 + x + 1)^2 - 13(x^3 - 1) = 7(x - 1)^2 .$$

a) Tenglamaning haqiqiy ildizlar soni nechta?

Javob: _____

b) Tenglamaning haqiqiy ildizlari ko'paytmasini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

37. Tenglamani yeching:

$$(\sin x + \cos x)^4 + (\sin x - \cos x)^4 + \sin 4x = 3 .$$

a) Eng kichik musbat qiymatini toping.

Javob: _____

b) Tenglama $[-\pi; 2\pi]$ oraliqda nechta ildizga ega?

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiiga ko'chirib yozing.

38. $f(x)$ funksiya uchun $2f(x) + f(1-x) = \frac{3x^2 + x + 4}{x(1-x)}$ tenglik o'rinli.

a) $f(2)$ ni toping.

Javob: _____

b) $f^{-1}(-2)$ ni toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

39. $y = 2\sqrt{x}$ funksiya berilgan.

a) Funksiyaga Ox o'qining musbat yo'nalishi bilan 30° burchak hosil qiladigan urinma o'tkazilgan. Urinmaning Ox o'qini kesgan nuqtasidan $y = 2\sqrt{x}$ funksiya uchun eng yaqin masofani toping.

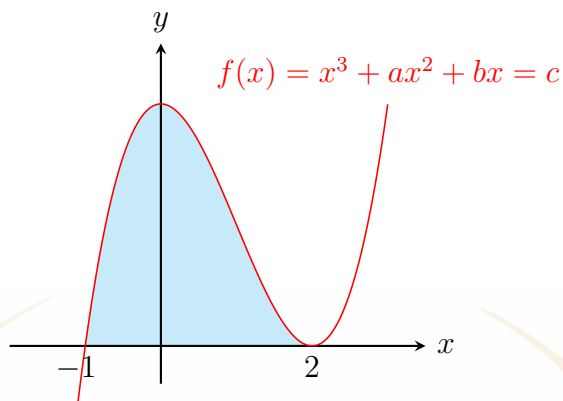
Javob: _____

b) $y = \frac{1}{2}x + 3$ funksiya uchun $y = 2\sqrt{x}$ funksiya uchun eng yaqin masofani toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

40. $y = x^3 + ax^2 + bx + c$ funksiyaning grafigi quyida tasvirlangan.



a) $a + b + c$ ni toping.

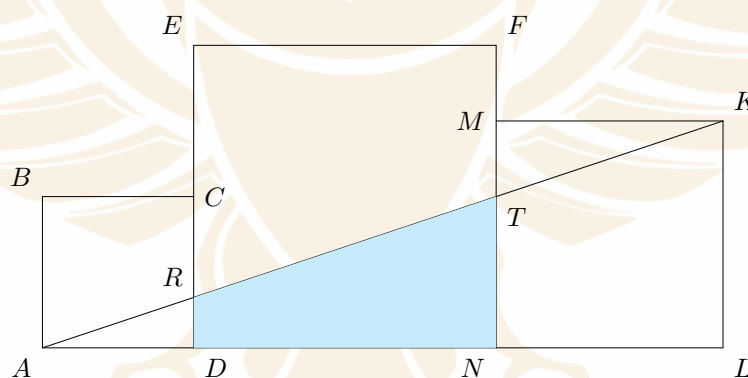
Javob: _____

b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

41. Quyida tomonlari 2,4,6 ga teng kvadratlar tasvirlangan.



a) Bo'yalgan soha yuzini toping.

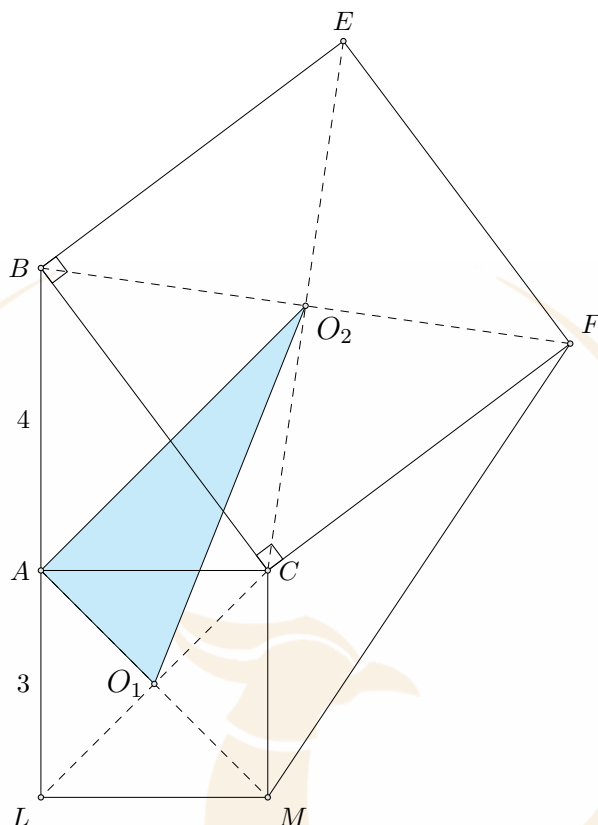
Javob: _____

b) TK kesma uzunligini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

42. ABC to'g'ri burchakli uchburchakning kateti va gipotenuzasi orqali tashqi kvadratlar yasalgan. $AB = 4$, $AL = 3$ bo'lsin.



a) FM ni toping.

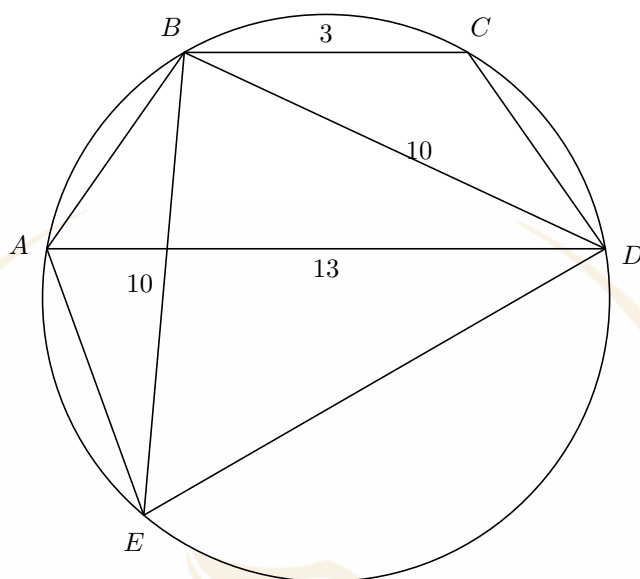
Javob: _____

b) Bo'yalgan soha yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqararasiga ko'chirib yozing.

43. Chizmada $ABCD$ trapetsiya berilgan. B nuqtadan aylanadagi E va D nuqtalargacha bo'lgan masofala 10 ga teng. A nuqtadan D nuqttagacha masofa 13 ga, trapetsiyaning kichik asosi 3 ga teng.



a) AE kesma uzunligini toping.

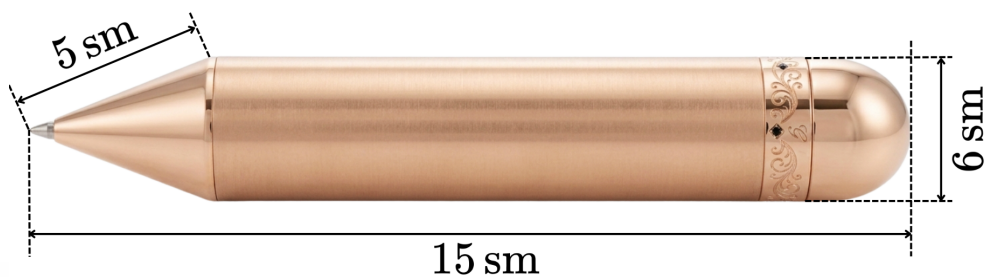
Javob: _____

b) $ABCDE$ beshburchak yuzini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasi ga ko'chirib yozing.

44. Quyida yarim shar, silindr va konusdan iborat jism tasvirlangan. Silindrning radiusi 3 cm ga, jismning umumiy uzunligi 15 cm ga teng bo'lsa,



a) Jismning to'la sirti yuzini toping.

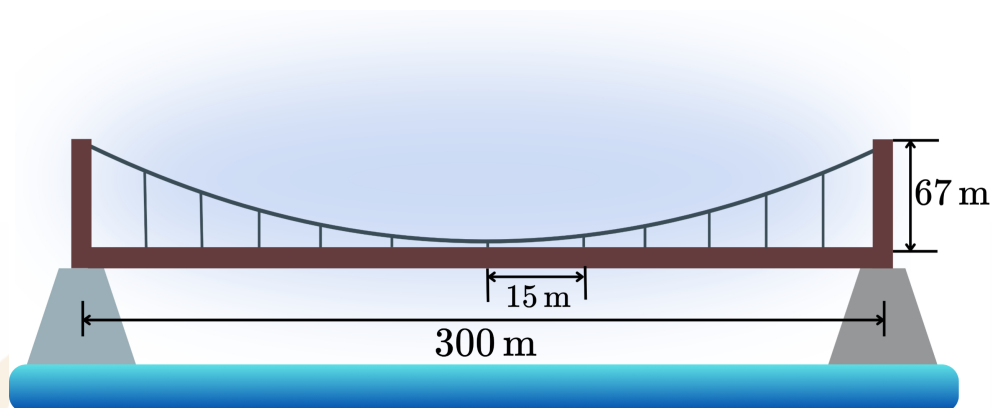
Javob: _____

b) Jismning hajmini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.

45. Quyida osma ko'prik tasvirlangan. Ko'prikning eng katta ustunlari orasidagi masofa 300 m, ustunlar orasidagi masofalar 15 m dan va katta ustun balandligi 67 m ga teng. Ko'prik markaziga koordinata ya'ni $O(0; 0)$ nuqta joylashtirilgan.



a) Parabola tenglamasini tuzing.

Javob: _____

b) 10 ta kichik ustunlarning uzunliklari yig'indisini toping.

Javob: _____

Diqqat! Javoblaringizni javoblar vaqaragasiga ko'chirib yozing.